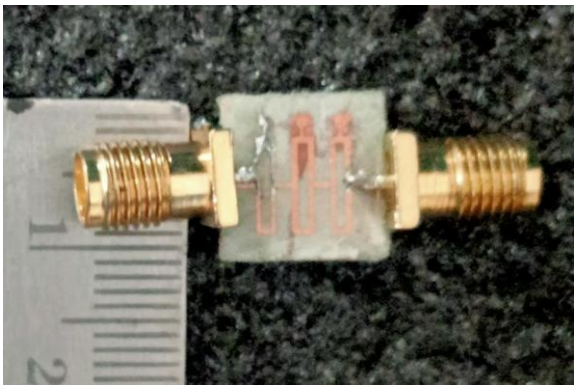
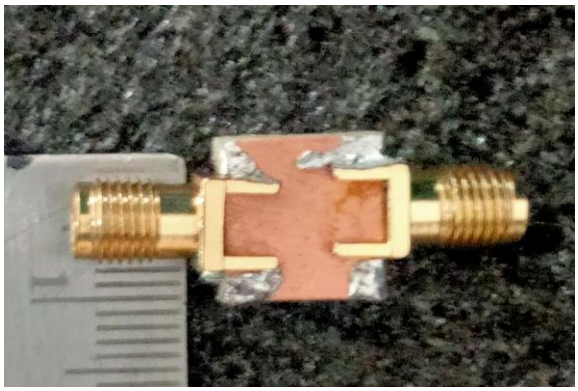
Fabrication & Testing Procedure of BPF Prototype



Frontend of prototype



Backend of Prototype

Step 1. Substrate Preparation

Selected dielectric substrate (FR4/Rogers).

Cleaned the surface to remove dust and oxidation.

Ensured proper thickness (h) for microstrip design**.**

Step 2. Layout Printing on Substrate

Printed the microstrip bandpass filter pattern on copper-clad substrate.

Used PCB etching method to remove unwanted copper.

Final structure contains:

Input transmission line

Resonator sections

Output transmission line

Step 3. Cutting & Finishing

Cut the substrate to required dimensions.

Smoothed edges to avoid unwanted radiation or reflections.

Step 4. SMA Connector Mounting

Two SMA connectors attached:

Left → Input

Right → Output

Soldered the center pin to microstrip line.

Ground of SMA connected to ground plane.

Step 5. Continuity & Basic Check

Checked with multimeter:

No short circuit between line and ground

Proper connection from input to output

Step 6. Measurement Setup

Used microwave analyzer (Keysight FieldFox).

Connected:

- Port 1 → Input SMA
- Port 2 → Output SMA

Step 8. Testing Procedure

Set frequency sweep range (e.g., 2.5GHz – 7.5GHz)**.**

Measured:

- S21 (Insertion Loss) → transmission response
- S11 (Return Loss) → reflection

Fig 2: Return Loss(S11)

Fig 3: Insertion Loss(S21)

Step 9. Obtained Results

i. Graph shows multiple peaks:

   Indicates resonant frequencies

ii. Passband identified at:

   Frequency where S21 is highest(maximum transmission).